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Beyond $\alpha = 0.05$

A Data-Driven System for Alpha Level Selection

Master's Thesis

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Hypothesis Testing

- Decisions in hypothesis testing are based on a simple rule:

$$\begin{aligned} \text{p-value} &\longrightarrow \text{Compare with } \alpha \\ p \leq \alpha &\Rightarrow \text{Reject } H_0 \\ p > \alpha &\Rightarrow \text{Fail to reject } H_0 \end{aligned}$$

- The choice of α directly affects the decision.

[Fis25]

Why do we use $\alpha = 0.05$?

- $\alpha =$ probability of a Type I error
- $\alpha = 0.05$ is commonly used as a standard threshold
- This choice is largely based on historical convention
- There is no clear justification for its universal use

[Fis25; CD82]

The Problem with $\alpha = 0.05$

One Size Fits All

Same α used regardless of

- sample size
- effect size

Cliff-Edge Effect

- $p = 0.049 \rightarrow$ Significant
- $p = 0.051 \rightarrow$ Not significant

Replication Crisis

- Many statistically significant findings fail to replicate.

[Ope15; WL16]

Existing Approaches for Choosing α

- Fixed α levels (0.10, 0.05, 0.01)
- Power-based approaches (sample size, effect size)
- Decision-theoretic approaches (error costs)
- Stricter thresholds ($\alpha = 0.005$)

$\Longrightarrow$ **All approaches still rely on a fixed α**

[Coh88; Ber85; Ben+18]

Limitations of Current Approaches

- Still rely on a fixed α
- No adaptation to study characteristics
- No practical system for selecting α

$\Longrightarrow$ **This motivates a data driven approach**

Proposed Approach

- Build a system to recommend a study-specific α

$$\text{Sample Size} + \text{Effect Size}$$
$$\downarrow$$
$$\alpha_{\text{baseline}} \text{ (ML, from data)}$$
$$\downarrow$$
$$\text{Context Adjustment (statistical rules)}$$
$$\downarrow$$
$$\text{Final } \alpha$$

[Coh88; Bak16; HFC19]

Research Question

Can a data-driven, context-sensitive approach improve α compared to $\alpha = 0.05$?

System Architecture

$$\alpha_{\text{final}} = \alpha_{\text{baseline}} \times \mathbf{k_1} \times \mathbf{k_2} \times \mathbf{k_3}$$

INPUT

Sample Size + Effect Size

(Known initially)

→

STAGE 1

Random Forest Regression

Trained on 664 replication studies MAE-optimized

→

STAGE 2

Context Adjustment

Risk tolerance(K1) Test direction(k2) Multiple testing(k3)

→

OUTPUT

Final α Recommendation

Clamped to 0.001 – 0.15 Ready to use

Data Source

- Replication studies (OSF)
- Across multiple research fields
- 664 hypothesis tests
- Includes:
  - Sample size
  - Effect size
  - p-values

[Cha25; ED21; Sot17; Sot22; Li21; Ope15]

Example Dataset Rows

Sample Size	Effect Size	p-value	Optimal α
120	0.35	0.041	0.052
85	0.28	0.063	0.061
241	0.30	0.048	0.074
150	0.22	0.071	0.066
96	0.40	0.032	0.049

- Each row represents one hypothesis test.

Optimal Alpha Definition

Replicated

Keep original α
(decision validated)

Failed

Adjust α: largest
$\alpha < p$-value
(correct decision)

Example

$p = 0.047 \rightarrow$ failed to replicate
Optimal $\alpha = 0.04$ (not 0.05)

Predicting Baseline α

Machine Learning Task

- Goal: predict α_{baseline}.
- Input features:
  - Sample size
  - Effect size
- Target variable: optimal α .

Algorithms Evaluated

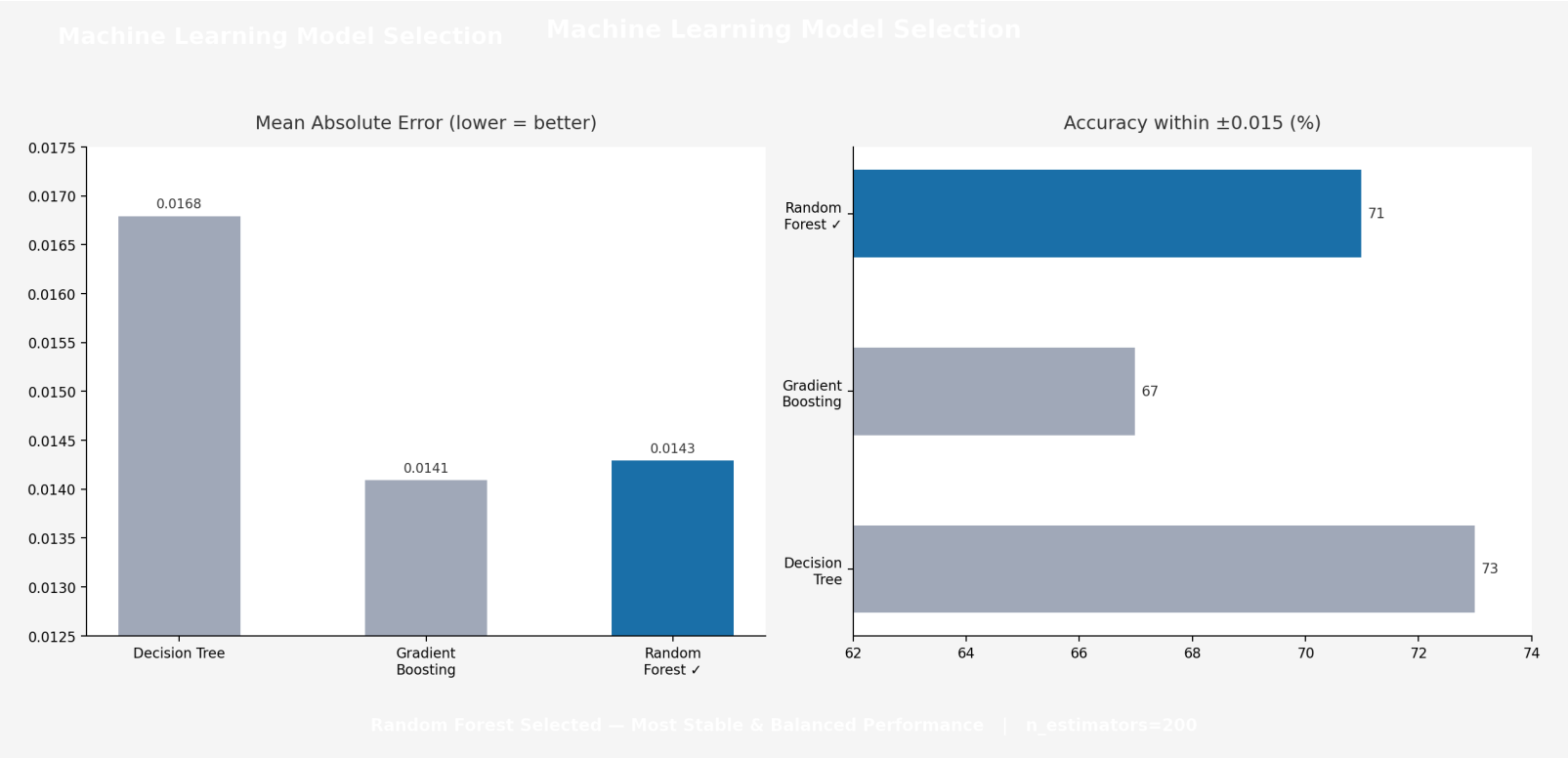
Regression Models Evaluated

- Decision Tree Regressor
- Gradient Boosting Regressor
- Random Forest Regressor

Evaluation Metric

Mean Absolute Error (MAE)

Machine Learning Model Selection



Random Forest achieves the lowest error

Selected Model: Random Forest

- Lowest prediction error
- Handles non-linear relationships.
- Stable performance

Implementation

Random Forest with **200 trees** used for baseline α prediction.

Stage 2: Context-Based Adjustment

Reason

Baseline α is not sufficient

Adjustment Factors

- Risk tolerance
- Test direction (one-tailed vs two-tailed)
- Number of hypothesis tests (multiple testing)

Adjustment Factors

k_1 — Risk Tolerance

- Very Strict $\times 0.50$
- Strict $\times 0.75$
- Neutral $\times 1.00$
- Lenient $\times 1.25$
- Very Lenient $\times 1.50$

k_2 — Test Direction

- Two-tailed $\times 1.00$
- One-tailed $\times 2.00$

Reflects rejection region allocation.

k_3 — Multiple Testing

Bonferroni correction:

$$\alpha/m$$

where m = number of hypothesis tests.

Final clamping: $0.001 \leq \alpha_{final} \leq 0.15$

Final Alpha Calculation

Adjusted Significance Level

$$\alpha_{final} = \alpha_{baseline} \times k_1 \times k_2 \times k_3$$

Interpretation

- $\alpha_{baseline}$: predicted by the machine learning model.
- k_1: adjustment for risk tolerance.
- k_2: adjustment for test direction.
- k_3: adjustment for multiple hypothesis testing.

System Deployment

Implementation

- Implemented in **Python**.
- ML model:**scikit-learn**.
- interface:**Streamlit**.

$$\text{User Inputs} \implies \text{Recommended } \alpha$$

System Interface

Evaluation Setup

Dataset

- Tested on 20 failed replication studies

Comparison

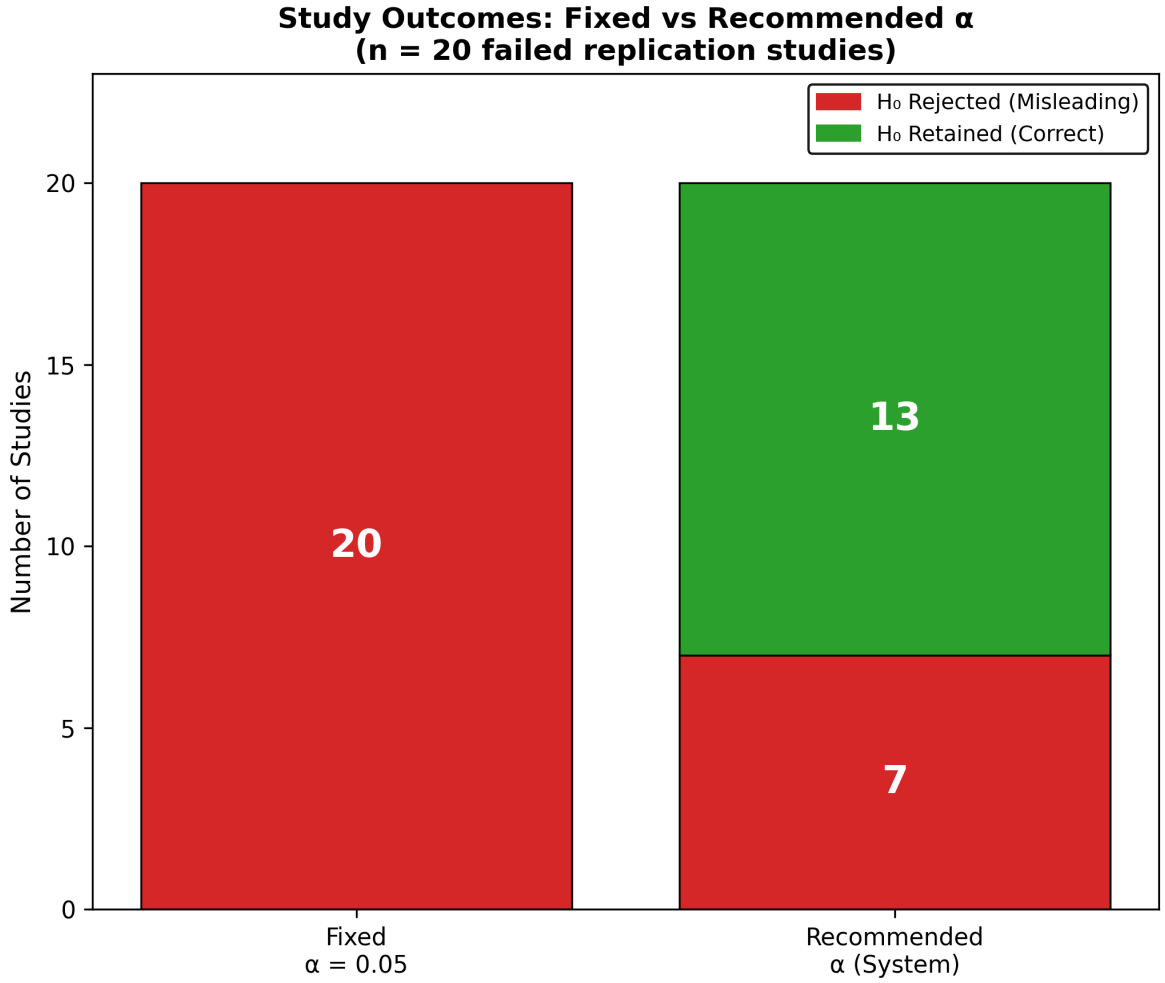
- $\alpha = 0.05$
- Recommended α

Evaluation

- Decision outcomes

$\Longrightarrow$ **Goal: Reduce incorrect rejections**

Main Findings



Reduces false positives by 65% (13 out of 20 cases)

Limitations

Key Limitations

- Limited dataset size
- Limited input features
- Dependent on available replication data
- Results provide guidance, not strict rules

Conclusion

Key Takeaways

- $\alpha = 0.05$ is not universally appropriate
- α should depend on study characteristics
- A data-driven approach improves α selection
- Combines machine learning with interpretable adjustments
- **Statistical significance should be adaptive, not fixed**

Future Work

Next Steps

- Expand replication dataset
- Include additional study features
- Explore alternative ML models
- Deploy as a fully accessible web application

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Vielen Dank

für Ihre Aufmerksamkeit

Literature

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